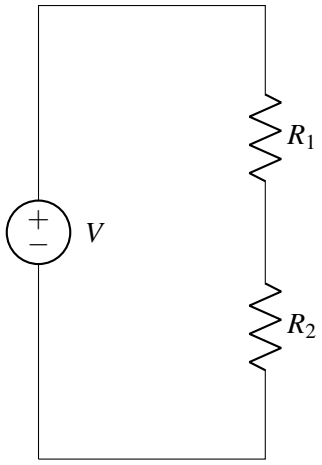
CSM
Fall 2026

EE 64
Worksheet 1

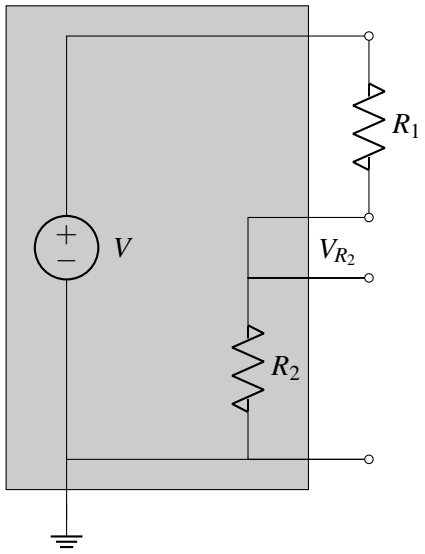
1. Voltage Divider Properties

Let's take a systematic look at the voltages across a resistor, and see how other components in the circuit can affect it. Consider the following circuit:



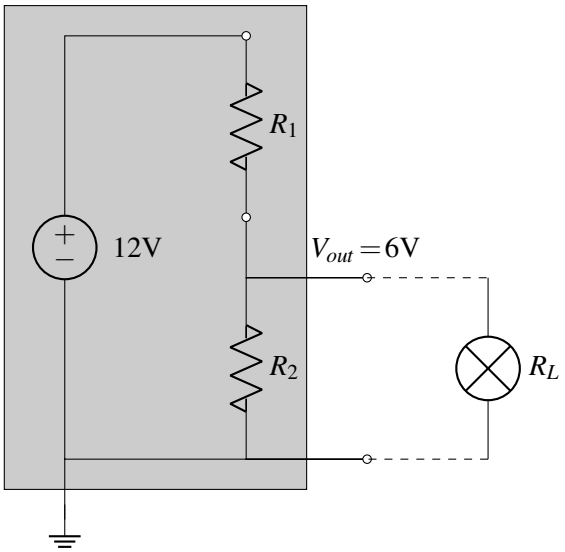
(a) Calculate the voltage drop across R_1 and R_2 using series resistance calculations.

(b) Suppose we want to manipulate the voltage across R_2, but it's locked in a box with the voltage source, as denoted below. Can we use R_1 to manipulate V_{R_2}? What range of voltages can we achieve?



(c) Now let's try using our new variable voltage source to power a light bulb with resistance R_L, where the threshold voltage for lighting the bulb is 6V. Find R_1 and R_2 so that the voltage across R_2 is this threshold voltage; that is, $V_{R_2} \equiv V_{out} = 6\text{V}$. Assume we have a 12V voltage source.

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(d) Now that we found an R_1 and R_2 that seem to divide our voltage source appropriately, let's try to connect the bulb to the ends of R_2. Remember, the bulb has a resistance R_L. Calculate the voltage across R_1, R_2 and the light bulb when it is connected. Will the light bulb turn on?

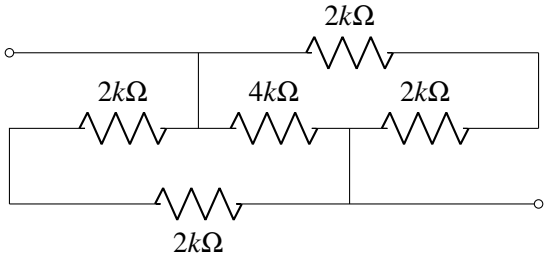
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2. Never fail at resistor equivalence

(a) Using what you learned about resistance above ($R = \frac{\rho \cdot l}{A}$), explain what would happen if you connect two resistors in series.

(b) Using the same resistance equation from part (a), explain what would happen if you connect two resistors in parallel.

(c) Now let's apply some of these ideas to a circuit. For the circuit below, mark all the nodes.



(d) Mark which nodes are 2-nodes and multi-nodes. 2 nodes are connected to only 2 components, and multi-nodes are connected to 3 or more components.
Note: *Whenever nodes are marked across which equivalent resistance must be found, those are considered 'components' because something could be connected there.*

(e) Resistors that are connected to 2-nodes are considered to be in series. Redraw the circuit, and find the 2 and multi-nodes again after combining the resistors connected to 2 nodes.

(f) Now we should be left with only multi-nodes. So far, we have seen that any resistors connected to 2-nodes are in series. We will see what happens to resistors connected to multi-nodes. Begin by writing out the 2 nodes that each resistor is connected to.

(g) If you have 2 or more resistors that are connected to the same 2 nodes, then they are in parallel. What does this mean for the 3 remaining resistors?

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3. Resistance is not Futile!

Learning Goal: Introduce the concept of resistivity and provide physical intuition for how changes in physical properties/dimensions of a resistor changes its resistivity. Read *Note 12 Section 12.3-12.4* to learn more.

Resistivity is a **physical property** of the material that quantifies how much it opposes the flow of electric current.

Assume that in an ideal case, the cross-section and physical composition of the wire are uniform, We can find its resistance with the equation below:

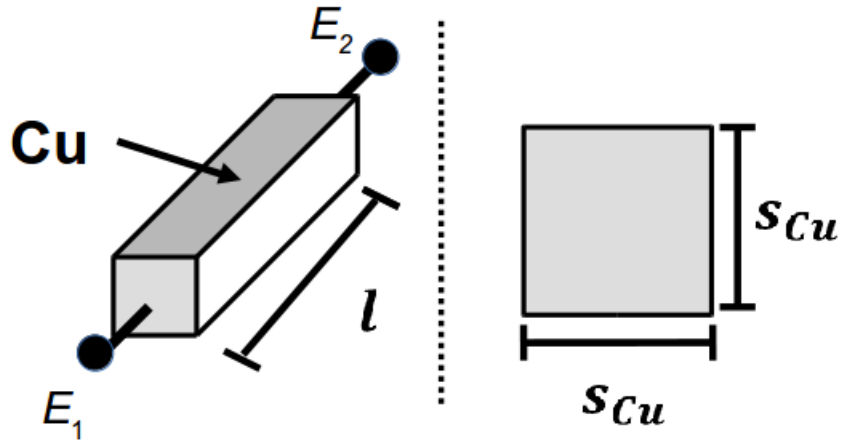
$$R = \rho \frac{L}{A}$$

Here, ρ stands for the resistivity of the wire, R stands for its resistance, A stands for the area of the cross section of the wire, and L stands for the length of the wire.

We will be frequently referencing some of the following variables:

- A: the cross section area of a single wire.
- L: the length of a single wire.
- ρ_{cu}: resistivity for the material copper.
- ρ_{Al}: resistivity for the material aluminum.

(a) A copper (Cu) structure with a square cross-section is shown below. Given the material parameters, calculate the resistance R_{Cu} of the structure between E_1 and E_2.

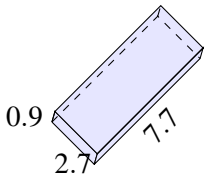


ρ_{Cu}	$1 \times 10^{-8}\,\Omega\,\text{m}$
s_{Cu}	5 nm
l	75 nm

(b) Now you are given a copper slab with dimensions 0.9cm, 2.7cm, and 7.7cm as denoted on the figure below. The dimensions and the resistivity of the slab remain the same throughout the rest of the problem.

Suppose we connect opposite faces of the slab to a voltage source so that current can flow through it. Notice that there are **3 possible configurations** we can have for this slab, leading to 3 possible directions in which the current can flow (drawn in the answer choices below). Which direction will

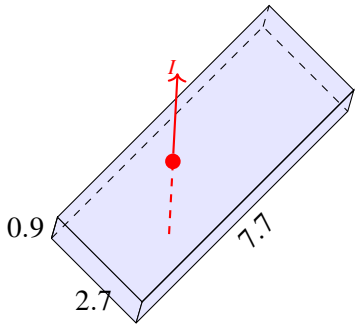
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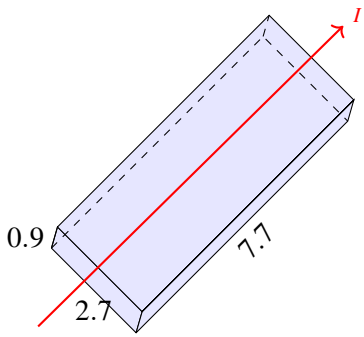
lead to the **highest** current flow I assuming we use the same voltage source all three times? Assume the resistivity is the same throughout the slab and does not vary with respect to

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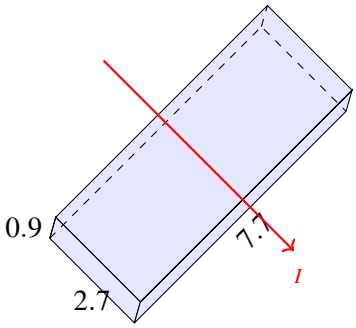
i. Parallel to 0.9 (current goes through 2.7, 7.7 face):



ii. Parallel to 7.7 (current goes through 0.9, 2.7 face):



iii. Parallel to 2.7 (current goes through 0.9, 7.7 face):



iv. Choices (i) and (iii) are tied for the maximum current flow.

v. The direction doesn't matter; the resistor is physically the same, so the current will be the same.

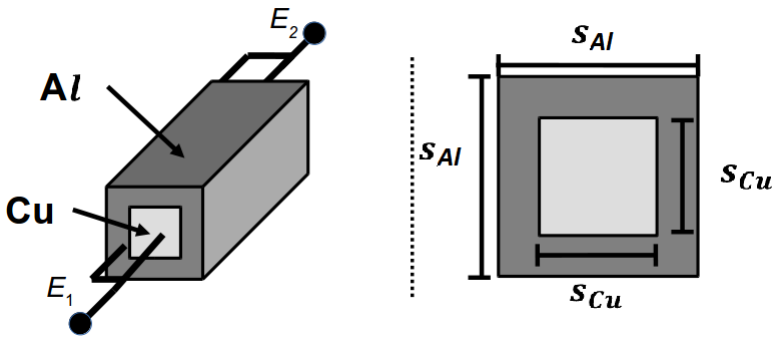
(c) Suppose we have N wires similar to the one described in part (a). We align them side by side to form a bundle of wires. Find the overall resistance of this bundle. Are the wires connected in series or parallel?

(d) **(PRACTICE)** Again how can we connect these N identical wires so that the equivalent resistance is the highest?

(e) **(PRACTICE)** Consider part (c) again, but this time, instead of N copper wires, we split the number evenly between aluminum wires and copper wires. We arrange $N/2$ copper and $N/2$ aluminum wires side by side, and push them to form a new bundle of wires. What is the overall resistance of this wire? (In terms of ρ_{cu}, ρ_{Al}, L, and A)

(f) Now consider this *core-shell nanowire* structure, where the outside shell made of Al and the inside made of Cu. E_1 and E_2 terminals are both connected to the faces of the Cu and Al structure, as shown in the picture. Draw a circuit diagram of the nanowire as a set of resistors, using R_{Al} for the resistance of the Al layer and R_{Cu} for the resistance of the Cu layer.

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(g) Based on your model from part (f) and the parameters given below, find the equivalent resistance R_{wire} between E_1 and E_2.

ρ_{Al}	$2 \times 10^{-8}\,\Omega\,\text{m}$
s_{Al}	$10\,\text{nm}$
ρ_{Cu}	$1 \times 10^{-8}\,\Omega\,\text{m}$
s_{Cu}	$5\,\text{nm}$
l	$75\,\text{nm}$

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